

The complete experiment record

9

RESEARCH AREAS

111

QUESTIONS + AUDITS

2,863

RECORDED RUNS

AREA		ENTRIES	REGISTER PAGES
Baseline comparisons		7	24-25
Step-size granularity		8	26-27
Training duration		3	28
Initial-rate robustness		5	29-30
Model and dataset scope		7	31-32
Proposed fixes		12	33-35
Meta-gradient measurements		17	36-40
Measurement checks		16	41-44
Mechanism and isolation		36	45-53

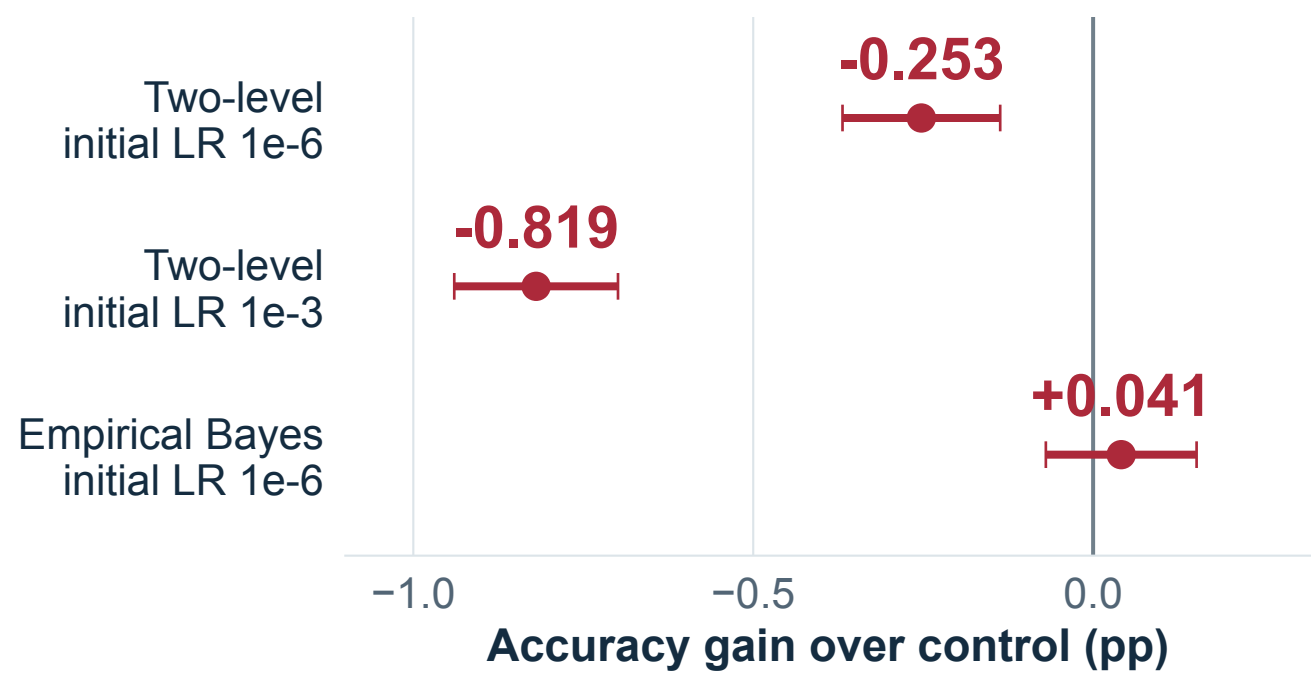
Accuracy: last 5 epochs. Bars: ±1 SE where available. pp = percentage points.

Green: goal met. Red: missed. Amber: mixed/open. Blue: corrected. Runs include reruns.

Hierarchy missed its accuracy goal

FAIL

GOAL	Improve accuracy by sharing step-size information.
A vs B	Two-level vs plain layerwise; empirical Bayes vs additive r=1
WHY	Shared information was expected to stabilize learning.



RESULT

BELOW BAR

Two-level loses.
EB gains too little.

EB: +0.04 pp
Success bar: +1.16

WHY FAIL? No variant meets the required improvement bar.

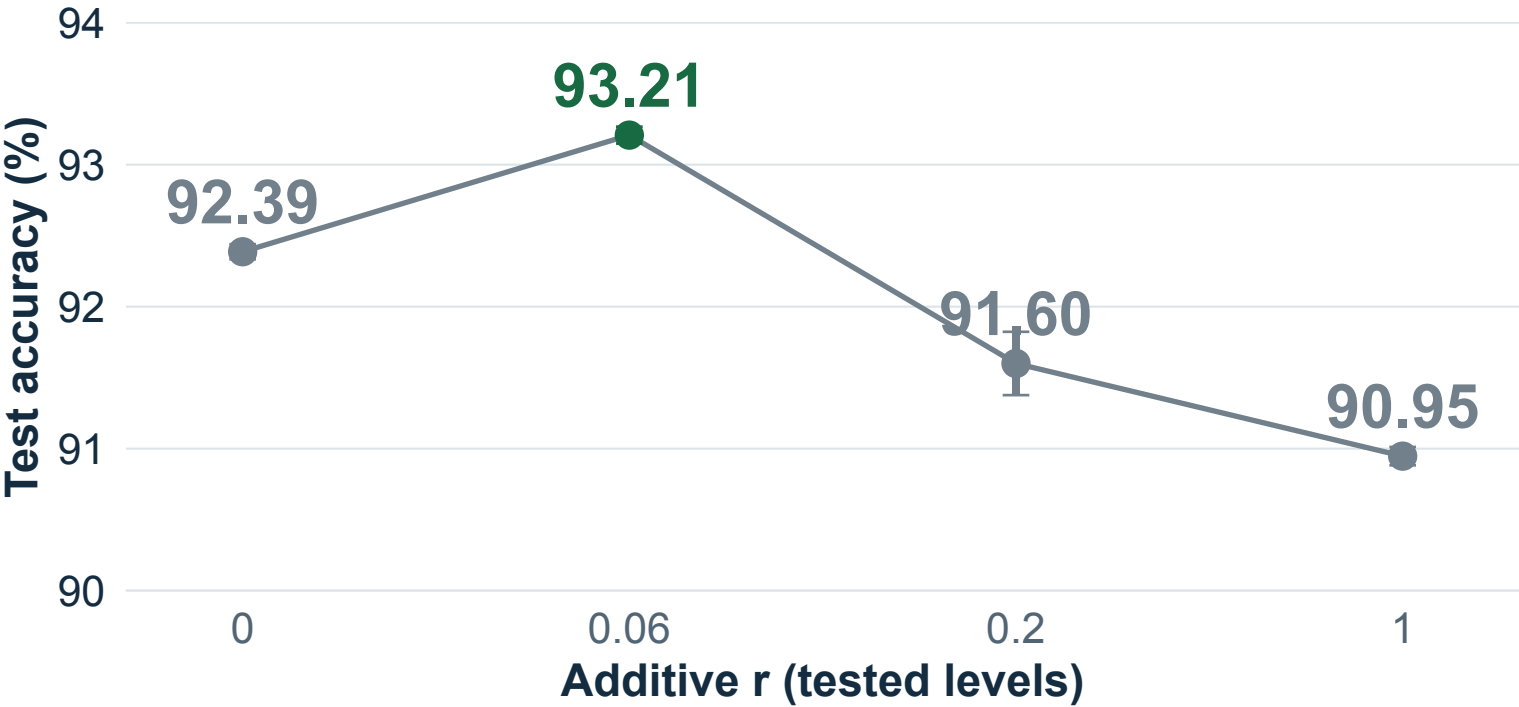
An additive benefit survives tuning

LOCAL WIN

GOAL Keep an interior-r benefit after meta-step tuning.

A vs B Best interior r vs best endpoint; both step-tuned

WHY Check whether the earlier gain was only a tuning artifact.



RESULT

+0.82 pp

Interior r wins after tuning.

Local result:
initial LR = 1e-6

WHY SUCCESS? Retuning the meta step does not erase the gain.

Several proposed fixes did not deliver

MIXED

GOAL

Improve MetaOptimize using structured fixes.

A vs B

Cosine priors / correlation grouping / pooling variants

WHY

Find an improvement beyond the original method.

TEST	RESULT	OUTCOME
Cosine prior	Missed its stopping-rule baseline	FAIL
Correlation-based grouping	Powered correlation contrast null	FAIL
M0 shrinkage	Tunable interpretation withdrawn	CORRECTED
Meta-gradient pooling	Weightwise rescue; layerwise flat	MIXED
Scaling law for r	Grid cannot identify a law	OPEN

WHY THIS RESULT?

No broadly validated fix emerged from these directions.

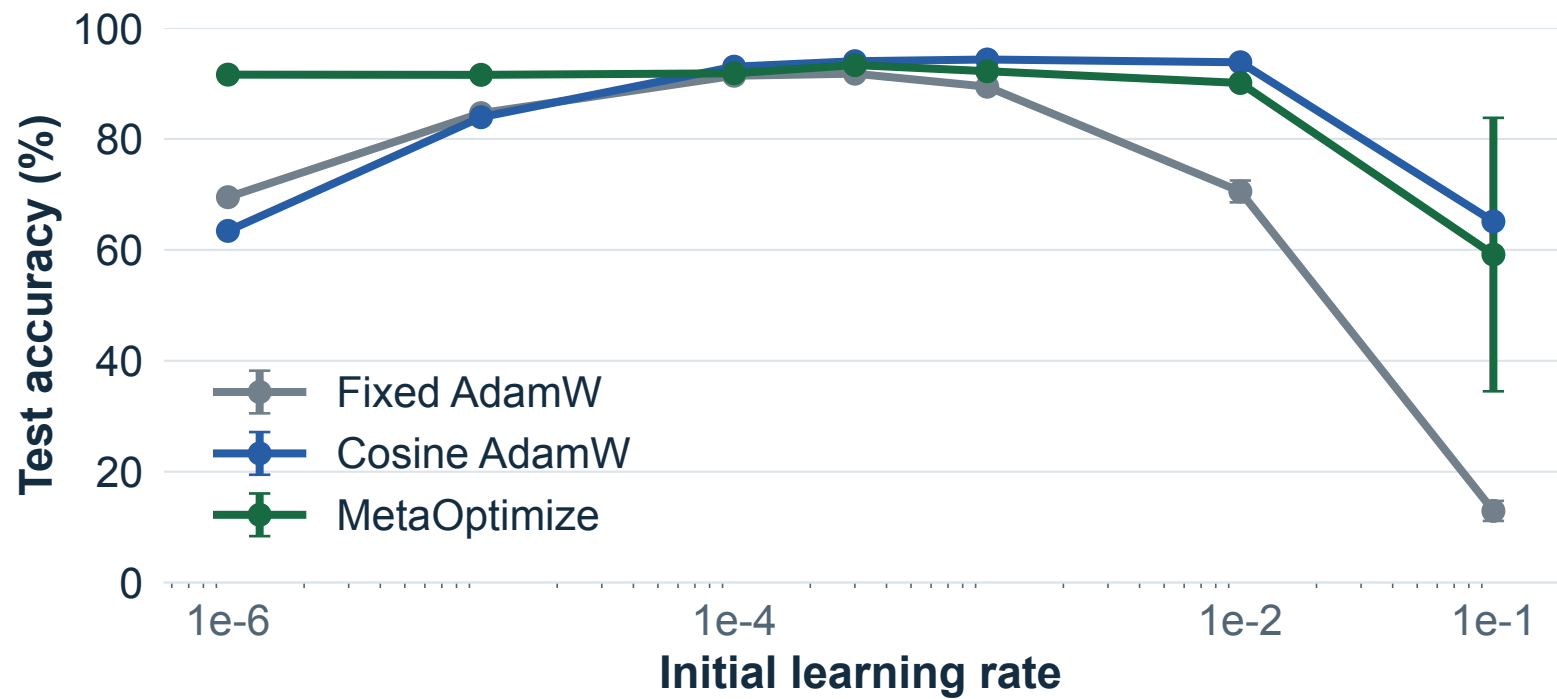
Initial-rate robustness: a real benefit

GRID WIN

GOAL Raise average accuracy across the tested rate grid.

A vs B MetaOptimize vs fixed-rate and cosine AdamW

WHY Measure how costly a poor initial rate can be.



RESULT

+3.16 pp

Mean over the same seven rates.

vs cosine AdamW
Cross-batch comparison

WHY SUCCESS? Learned rates improve this grid-average score.

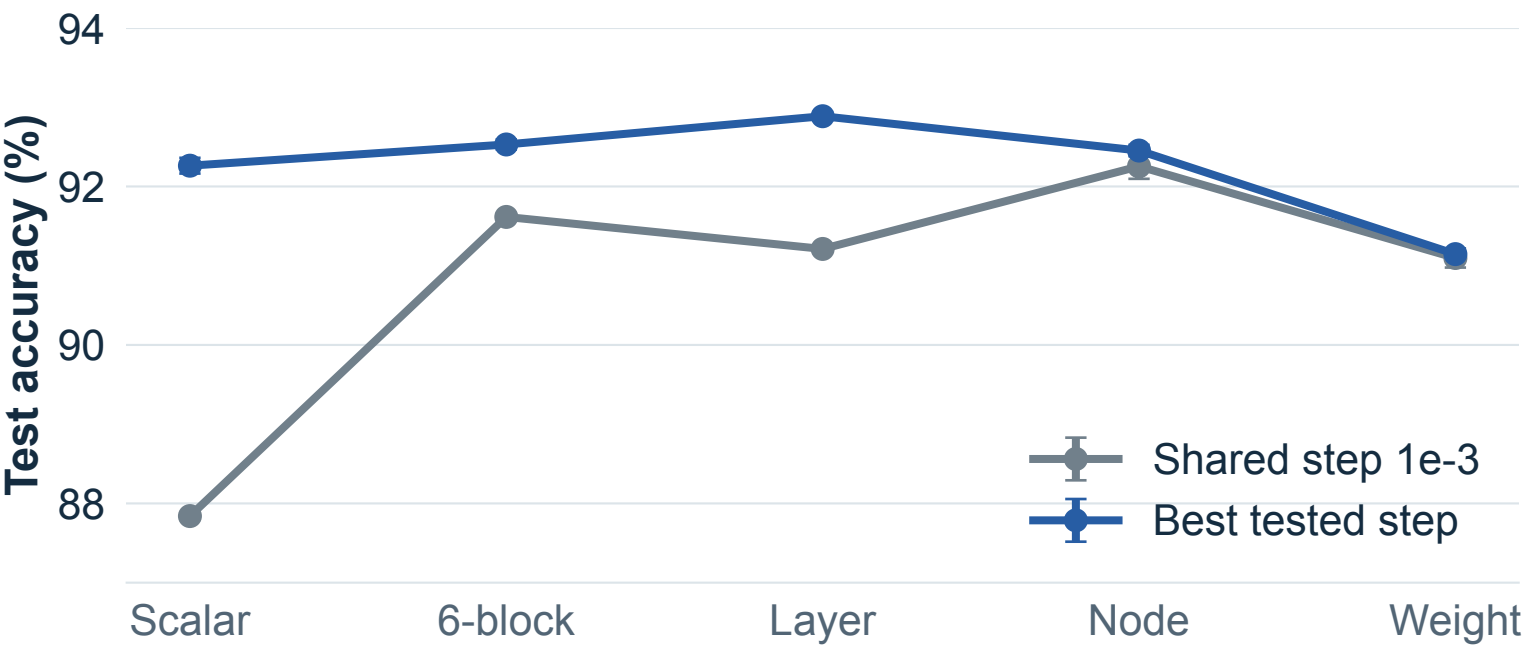
Layerwise peaks; weightwise trails

FAIL

GOAL Improve accuracy by making step-size groups finer.

A vs B Scalar / 6-block / layerwise / nodewise / weightwise

WHY Test whether more step sizes keep helping.



RESULT

-1.74 pp

Weightwise trails layerwise at peaks.

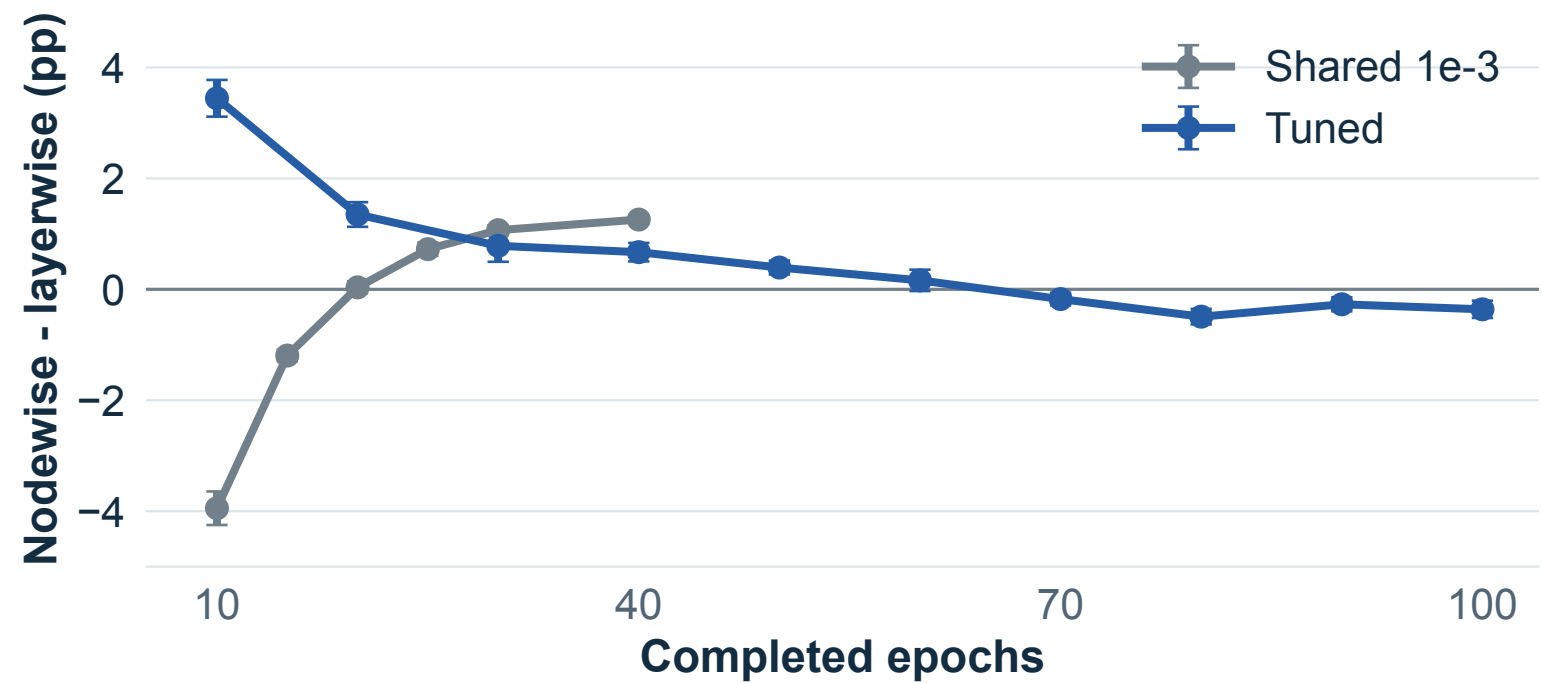
Observed grid peaks
Unequal, pooled cells

WHY FAIL? The finest partition is not the best-performing one.

The old horizon reversal fails retuning

FAIL

GOAL	Reproduce the claimed reversal after tuning.
A vs B	Nodewise minus layerwise: shared vs tuned meta steps
WHY	Separate a training-duration effect from poor tuning.



RESULT

-0.36 pp

Tuned difference at 100 epochs.

Inside the +/- 0.50 pp materiality band

WHY FAIL? The original direction does not survive retuning.

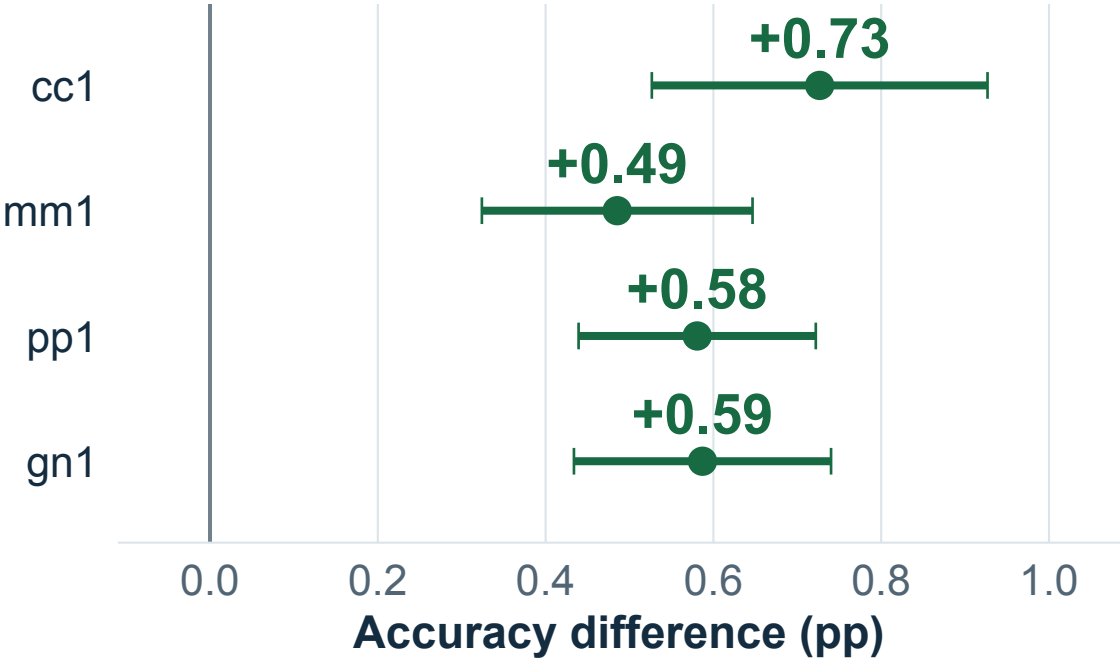
Partition choice matters at similar counts

LOCAL WIN

GOAL Improve accuracy while keeping group counts almost equal.

A vs B chunk777 (14,421 groups) vs nodewise (14,420)

WHY Test membership beyond the number of learned step sizes.



RESULT

+0.49–0.73

pp across four submissions.

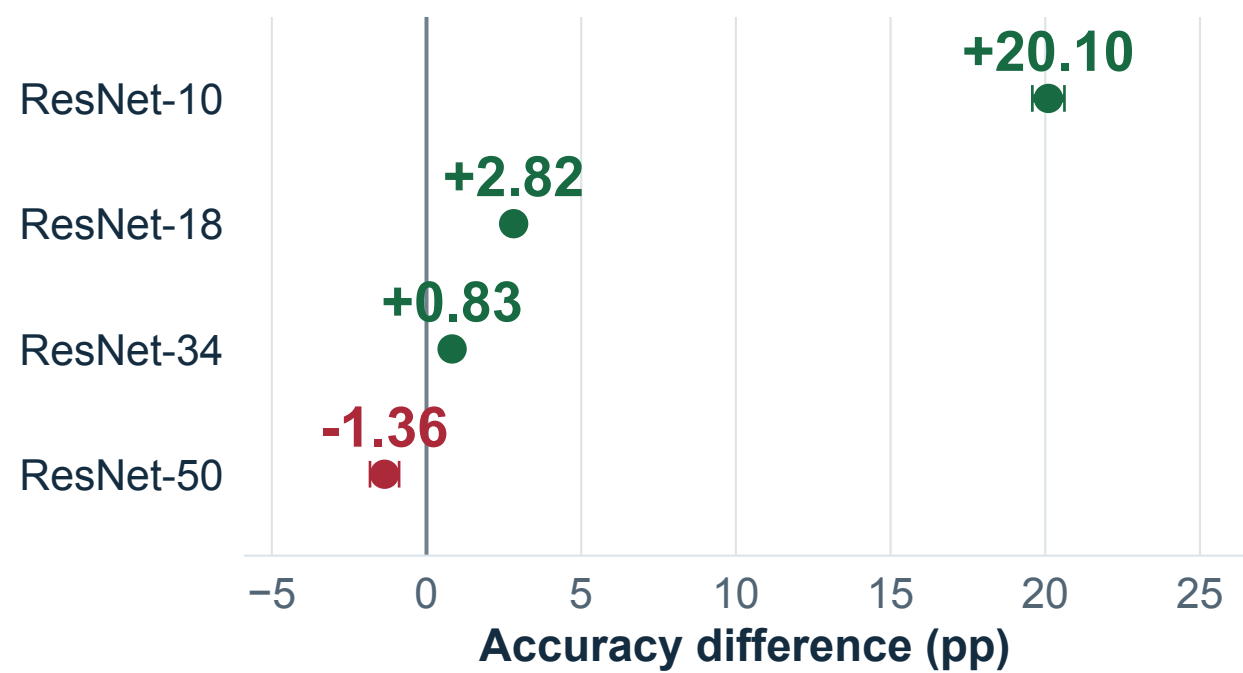
Overlapping seeds
Only 2 disjoint seed sets

WHY THIS RESULT? The chunk partition wins in all four submissions.

Model scale changes the size and sign

MIXED

- GOAL
- Transfer granularity gains across model scale.
- A vs B
- Plain layerwise vs scalar within each model
- WHY
- Check whether the ResNet-18 result generalizes.



RESULT

-1.36 pp

ResNet-50:
layerwise - scalar

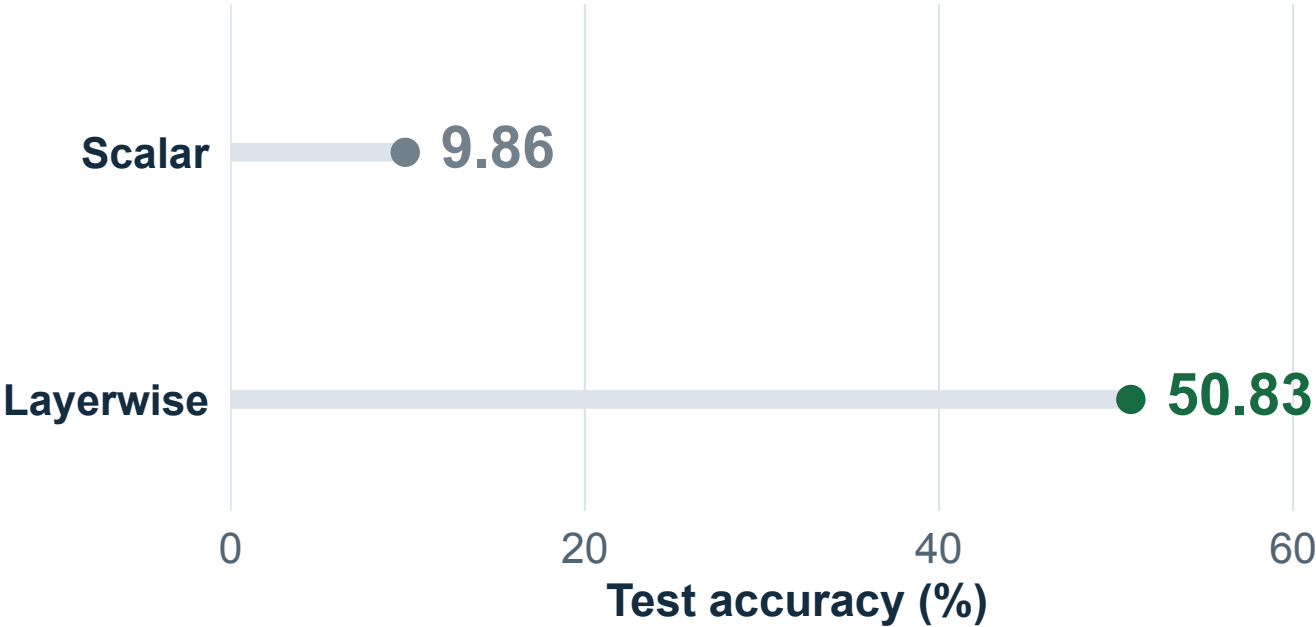
Initial rate 1e-6
Shared meta step 1e-3

WHY THIS RESULT? The same setting helps some models and hurts ResNet-50.

The granularity gap reaches Tiny-ImageNet

SUCCESS

- GOAL Replicate the granularity gap on a third dataset.
- A vs B Layerwise vs scalar on Tiny-ImageNet
- WHY Check whether the CIFAR result extends to new images.



RESULT

+40.98 pp

A third dataset shows a large gap.

Shared meta step
Target tuning untested

WHY SUCCESS? Layerwise recovers accuracy in the tested Tiny-ImageNet cell.

ImageNet-489: one win, two failed gates

MIXED

GOAL

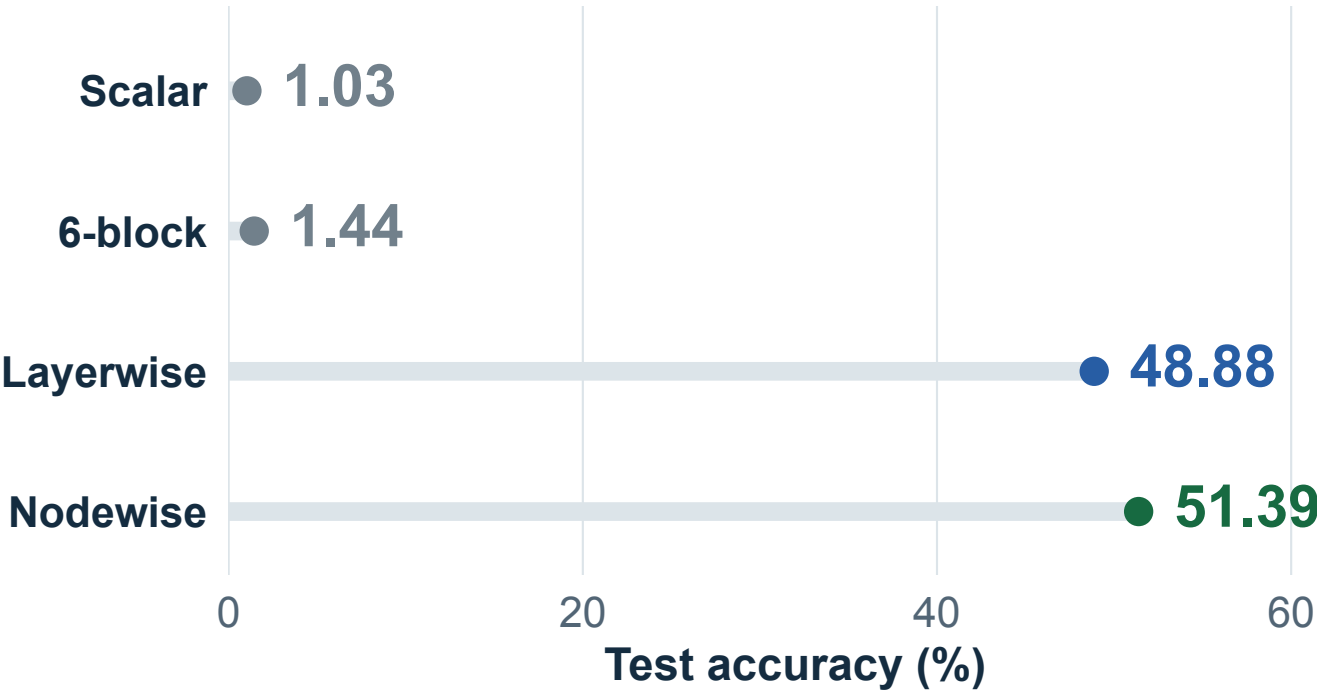
Replicate granularity effects on larger images.

A vs B

Scalar / 6-block / layerwise / nodewise on ImageNet-489

WHY

Test whether the result survives a larger dataset.



NODE: SUCCESS

+2.51 pp

Nodewise beats layerwise.

SCALAR / BLOCK: OPEN

Both fail the 5% floor gate

WHY MIXED?

Fine arms learn; scalar and block arms fail the floor gate.

Adaptation changes effective independence

MEASURED

GOAL

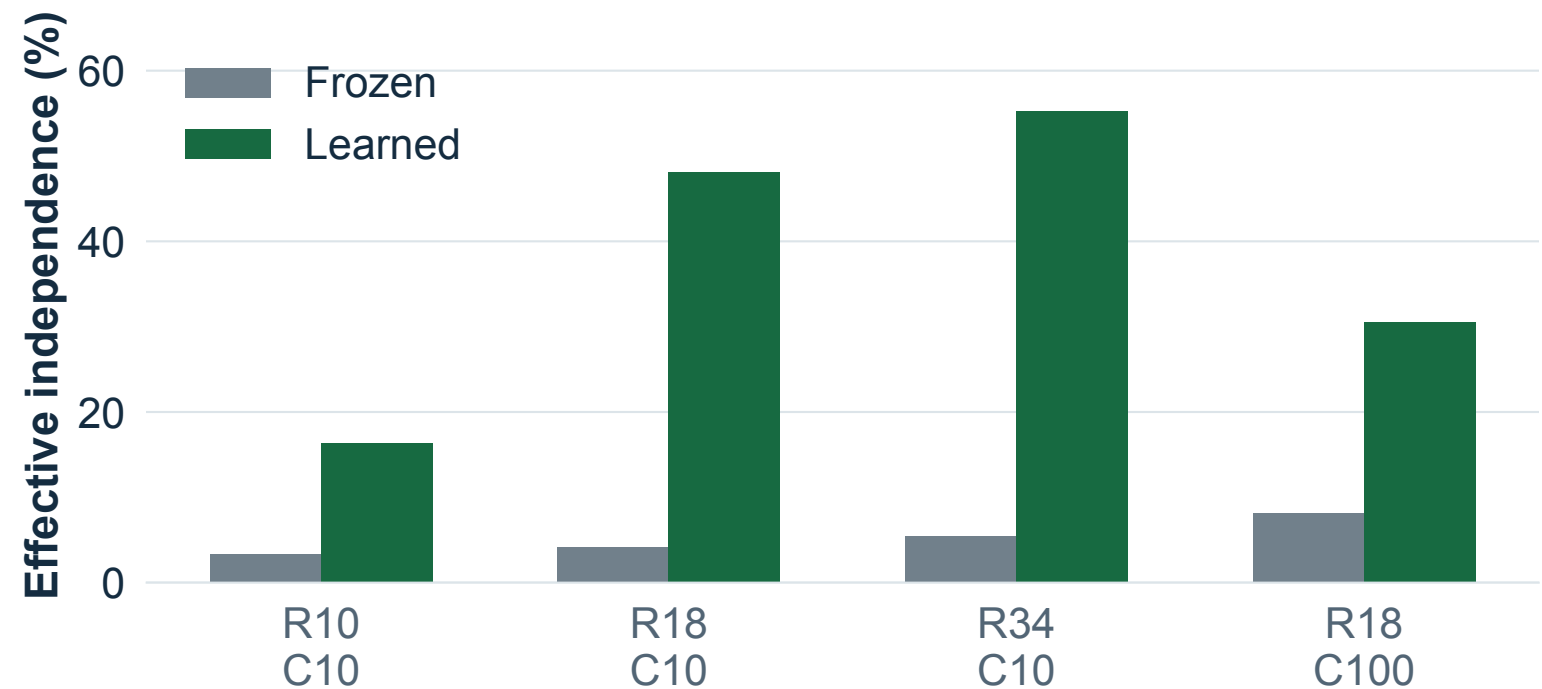
Measure dependence after the step sizes adapt.

A vs B

Frozen vs learned step sizes in four model/data families

WHY

Test whether millions of coordinates behave independently.



RESULT

16-55%

Effective fraction with learned rates.

20 epochs; SGDm
N_eff / m statistic

WHY THIS RESULT?

Adaptation reduces the dependence between coordinates.

A fine-scale field pattern repeats

MEASURED

GOAL

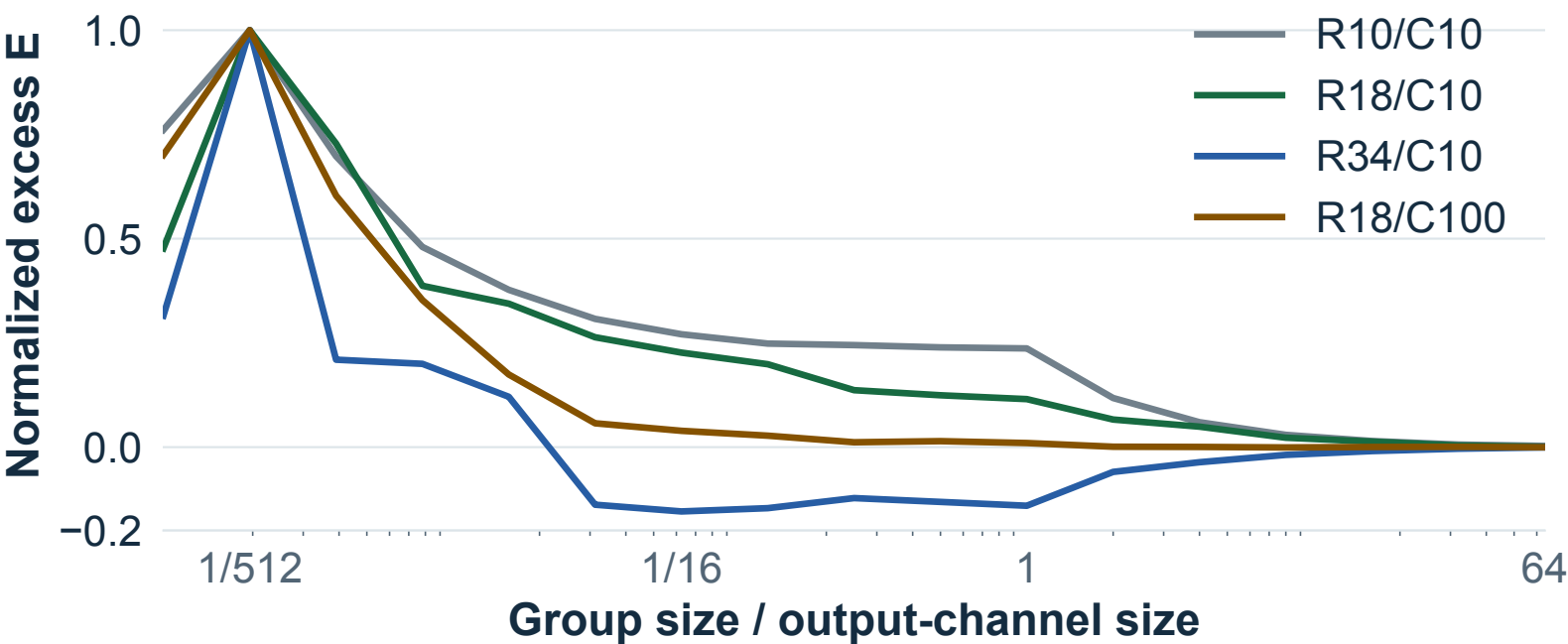
Locate a repeatable scale in marginal-sign structure.

A vs B

Observed blocks vs size-matched, within-tensor shuffles

WHY

Look for a structural guide to partition design.



RESULT

1/512

24 of 28 clean runs peak here.

4 peaks hit the grid edge
Their locations are bounds

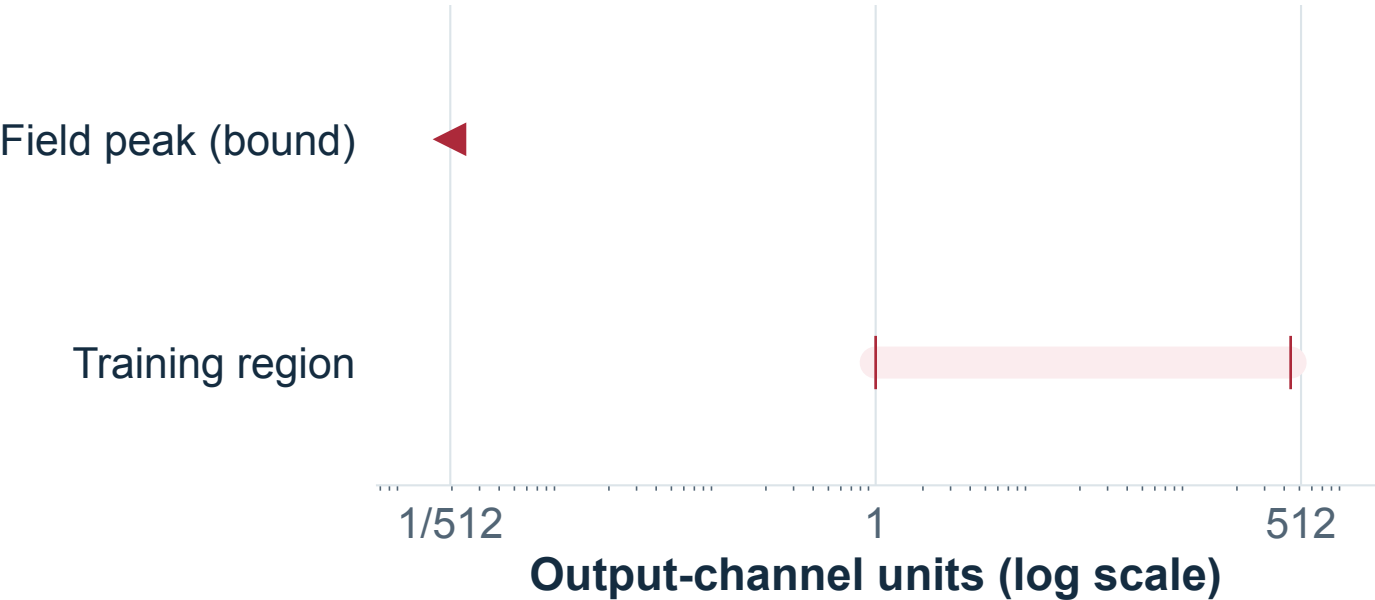
WHY THIS RESULT?

Peak location repeats; peak existence alone is mechanical.

Field structure did not predict the partition

FAIL

- GOAL Predict a useful partition from the measured field.
- A vs B Field-peak scale vs the best training-accuracy region
- WHY Turn a measurement into a working optimization method.



RESULT

$\geq 512\times$

Separation between the two scales.

At least 2.71 decades
Corrected historical bound

WHY FAIL? The measured scale does not locate the best training region.

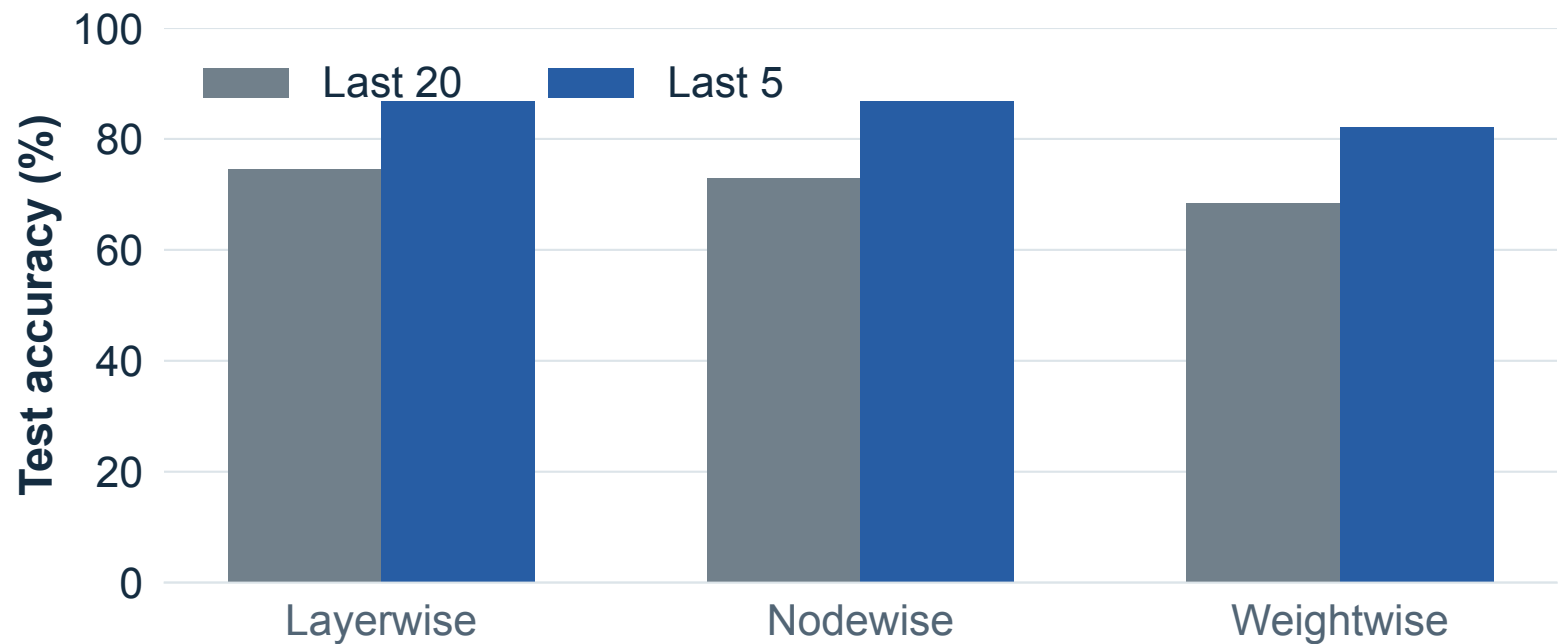
The measurement window changed the story

CORRECTED

GOAL Compare training budgets using the same tail metric.

A vs B Last 20 epochs vs last 5 epochs in a 20-epoch run

WHY Keep startup speed separate from endpoint accuracy.



RESULT

+13.3 pp

Mean shift in these three example cells.

Same runs
Different time window

WHY CORRECTED? A whole-run average had been presented as a tail average.

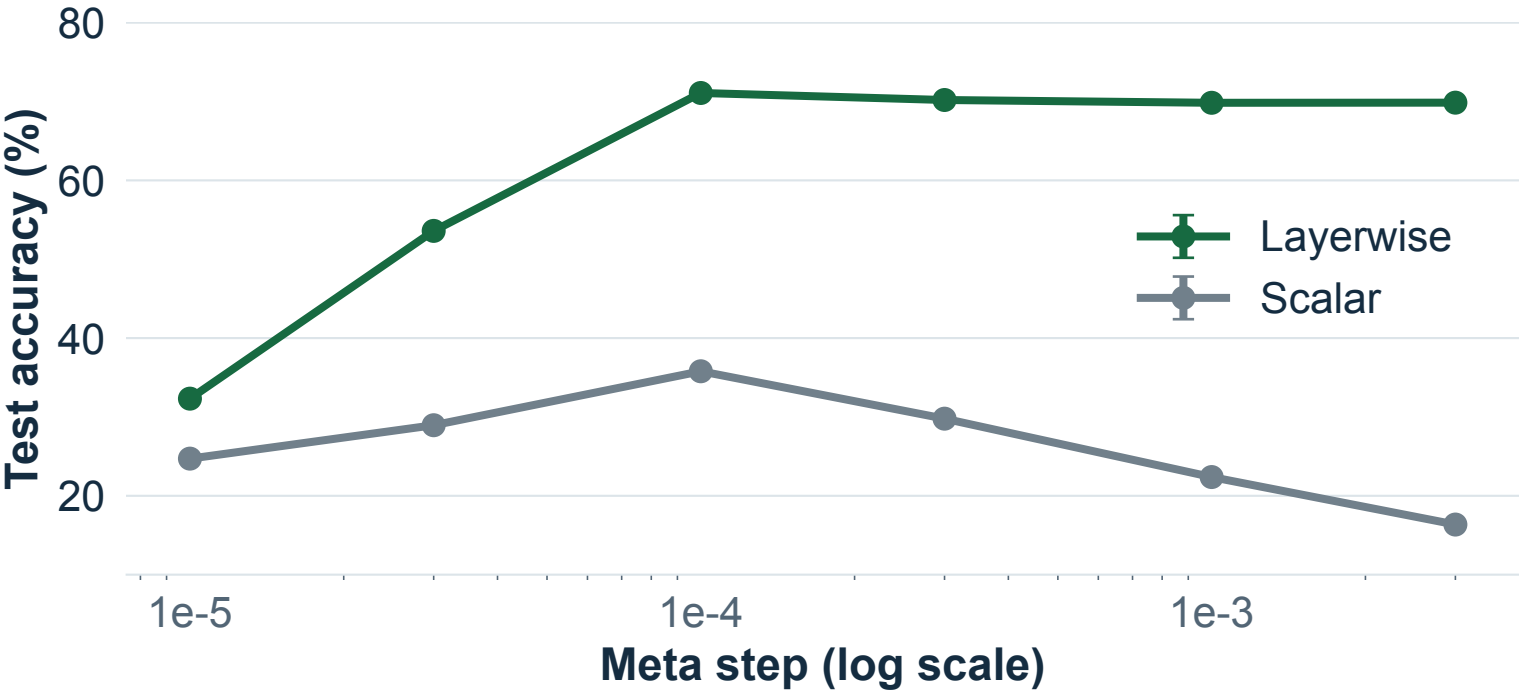
Granularity still matters after tuning

SUCCESS

GOAL Verify the accuracy gap after fair meta-step tuning.

A vs B Layerwise vs scalar; each at its best tested step

WHY Rule out a poor shared meta-step as the whole explanation.



RESULT

+35.29 pp

Layerwise keeps a large lead.

74.36% of the shared-step gap stays

WHY SUCCESS? Tuning shrinks the gap; most of it remains.

Mechanism tests eliminated several accounts

MIXED

GOAL

A vs B

WHY

Explain where the granularity gap comes from.

Head / sign dynamics / named drivers / reduction

Combine interventions with recorded training dynamics.

TEST	RESULT	OUTCOME
Classifier-head account	Head captures only 4.12% of gap	FAIL
Sign-disagreement peak	Peak elsewhere; correlation -0.046	FAIL
Named conv/head drivers	Pinned votes: three BN scales	FAIL
Element-count majority	Required statistic was not stored	OPEN
Reduction composition	One clean pair changes accuracy	MIXED

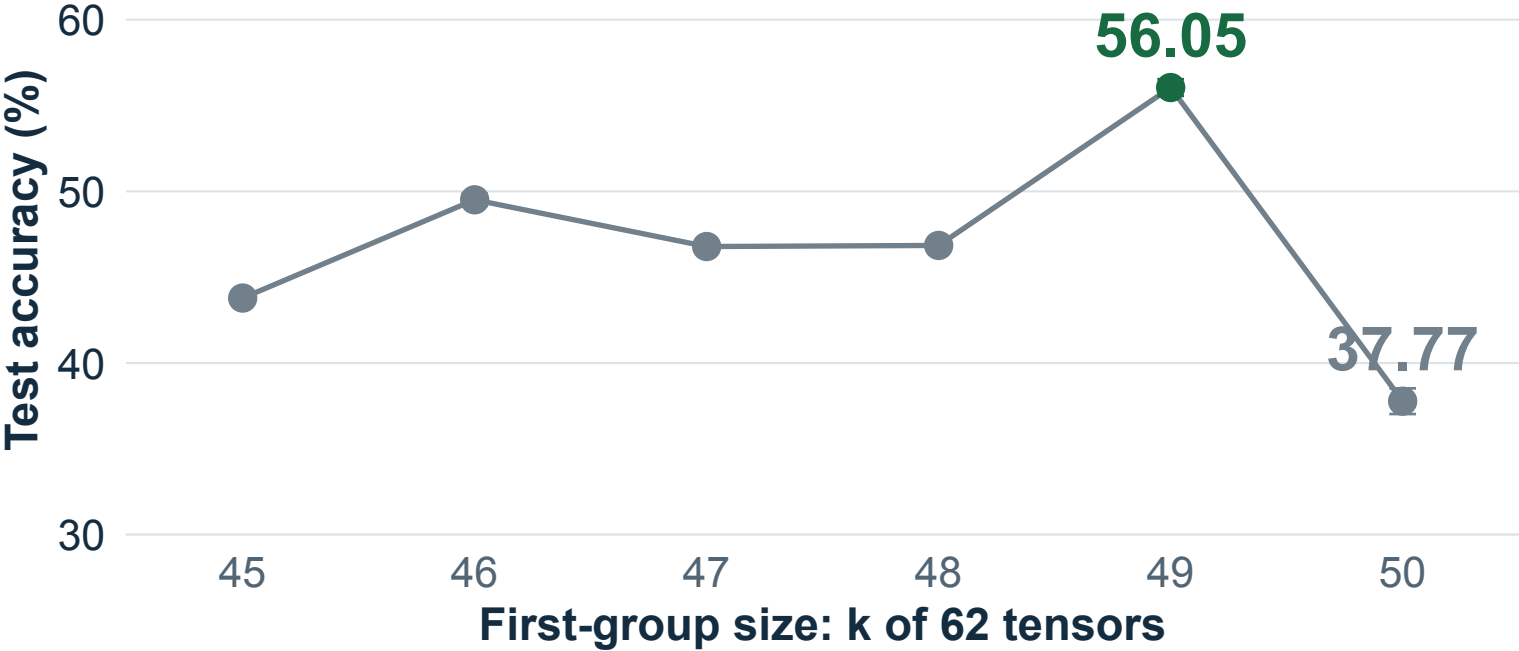
WHY THIS RESULT?

Several accounts fail; the causal explanation remains open.

One tensor boundary changes accuracy

LOCAL WIN

- GOAL**Locate a useful two-group split.
- A vs B**Cut after tensor 49 vs cut after tensor 50
- WHY**A sharp accuracy drop can nominate a target for intervention.



RESULT

18.27 pp

Lost by moving one tensor.

Best tested cut: 49
Mechanism still open

WHY LOCAL?

The useful boundary is located; its cause is open.

A tiny subset rescues ResNet accuracy

SUCCESS

GOAL

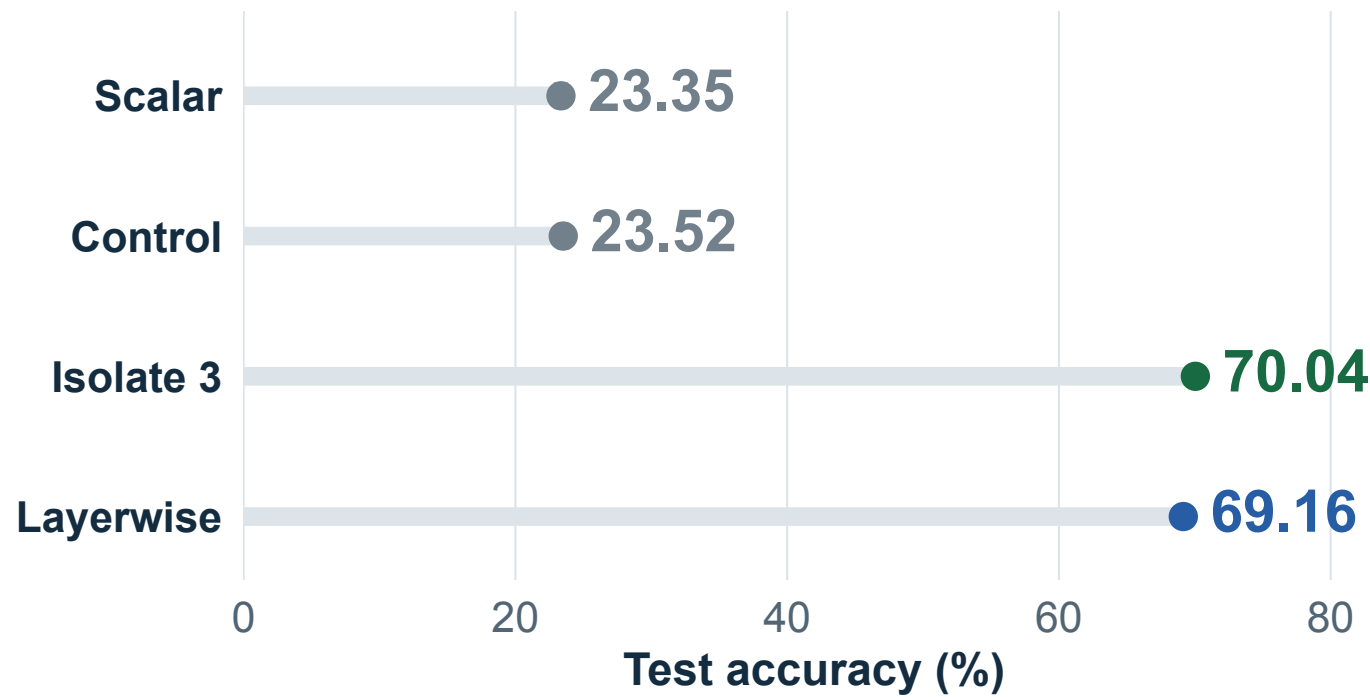
Recover layerwise-level accuracy with two groups.

A vs B

Three chosen BN scales vs matched late-layer control

WHY

Test the nominated tensors by changing their grouping.



RESULT

+46.52 pp

Above the matched control; two groups.

1,536 parameters

0.0137% of the model

WHY SUCCESS?

Chosen scales rescue accuracy; matched controls do not.

Rescue holds; pinning stays unresolved

MIXED

GOAL

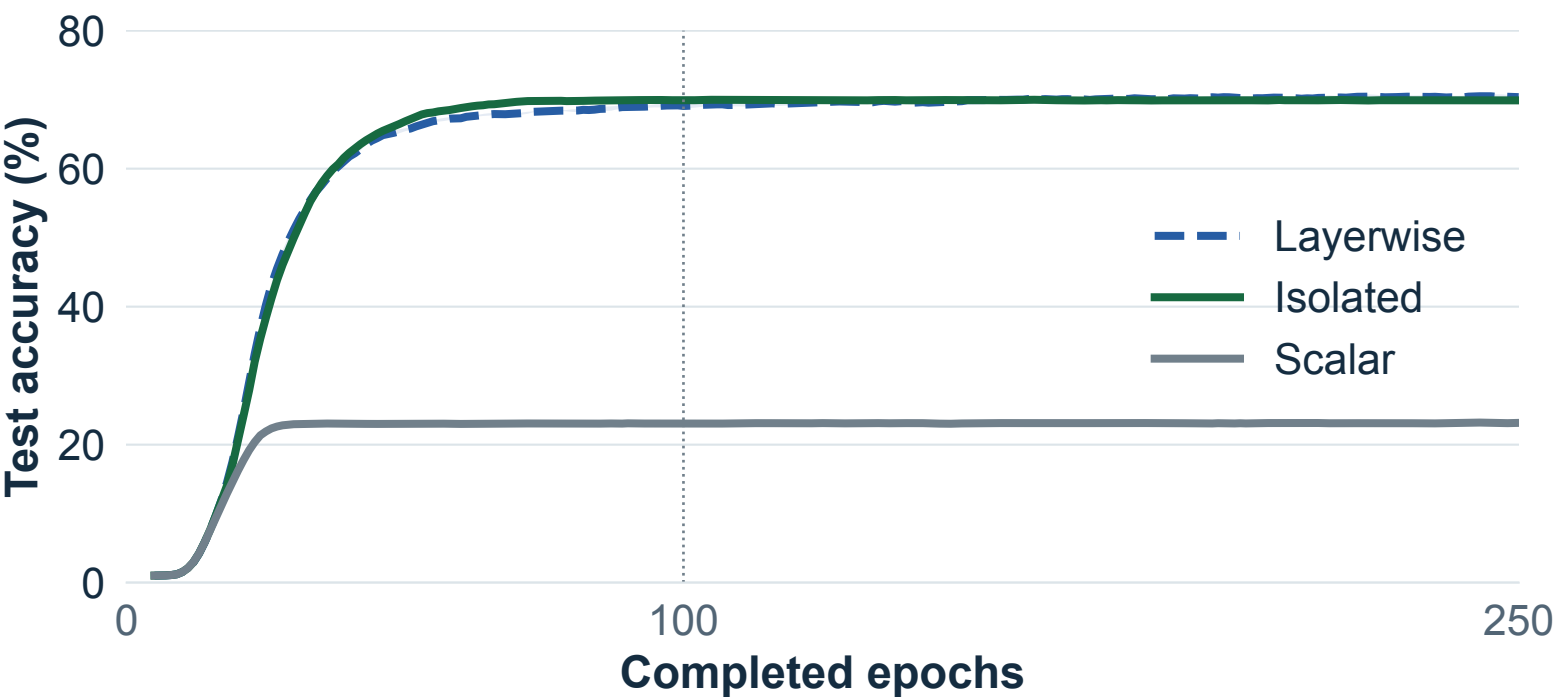
Check long-run rescue and the strict pinning test.

A vs B

Same ResNet runs: 100 vs 250 epochs

WHY

Test whether the rescue survives longer training.



ACCURACY: SUCCESS

99.77%

Isolation-minus-scalar gap retained.

PIN TEST UNRESOLVED

Exact-floor gate unmet

WHY MIXED?

Accuracy persists; the strict pin criterion is unmet.

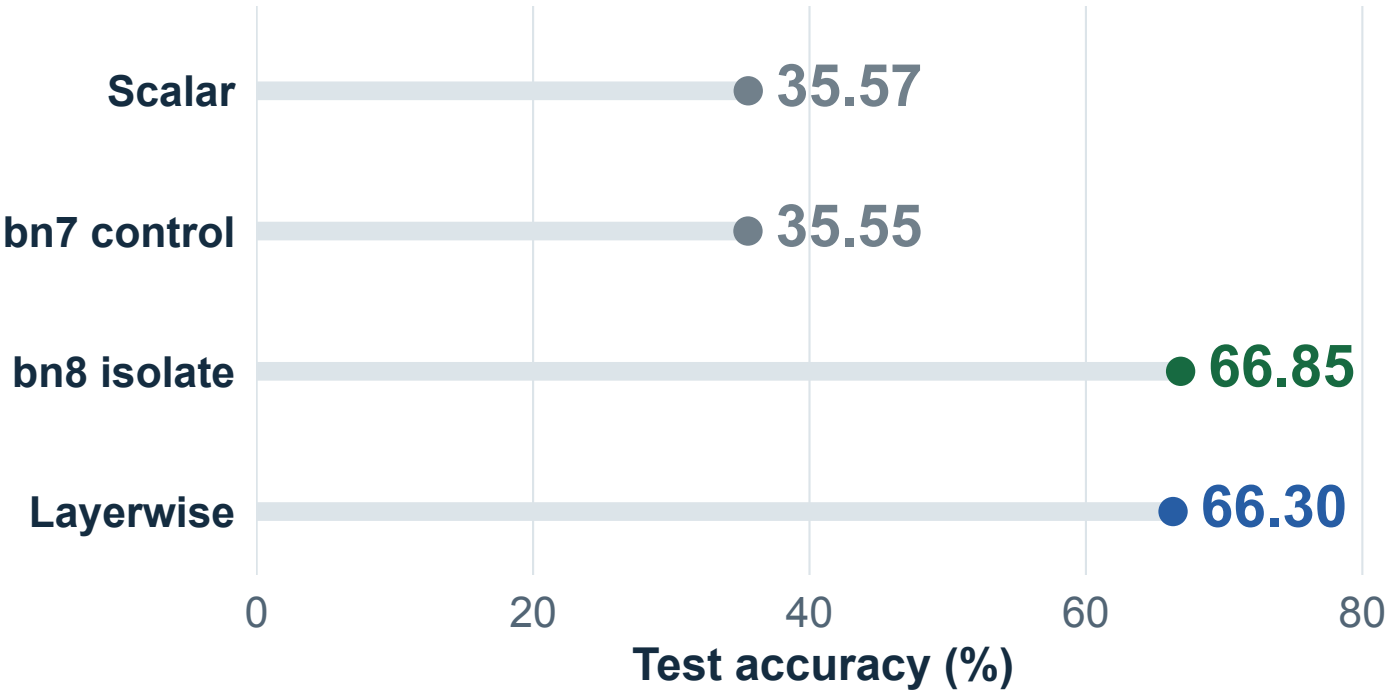
The rescue repeats in VGG

SUCCESS

GOAL Reproduce the rescue in another architecture.

A vs B Isolate bn8 vs matched bn7 in VGG11_bn

WHY Check whether the effect extends beyond ResNet.



RESULT

+31.30 pp

A second architecture.

One selected tensor
512 parameters

WHY SUCCESS? bn8 rescues accuracy; matched bn7 does not.

VGG cut location: completed test

UNRESOLVED

GOAL

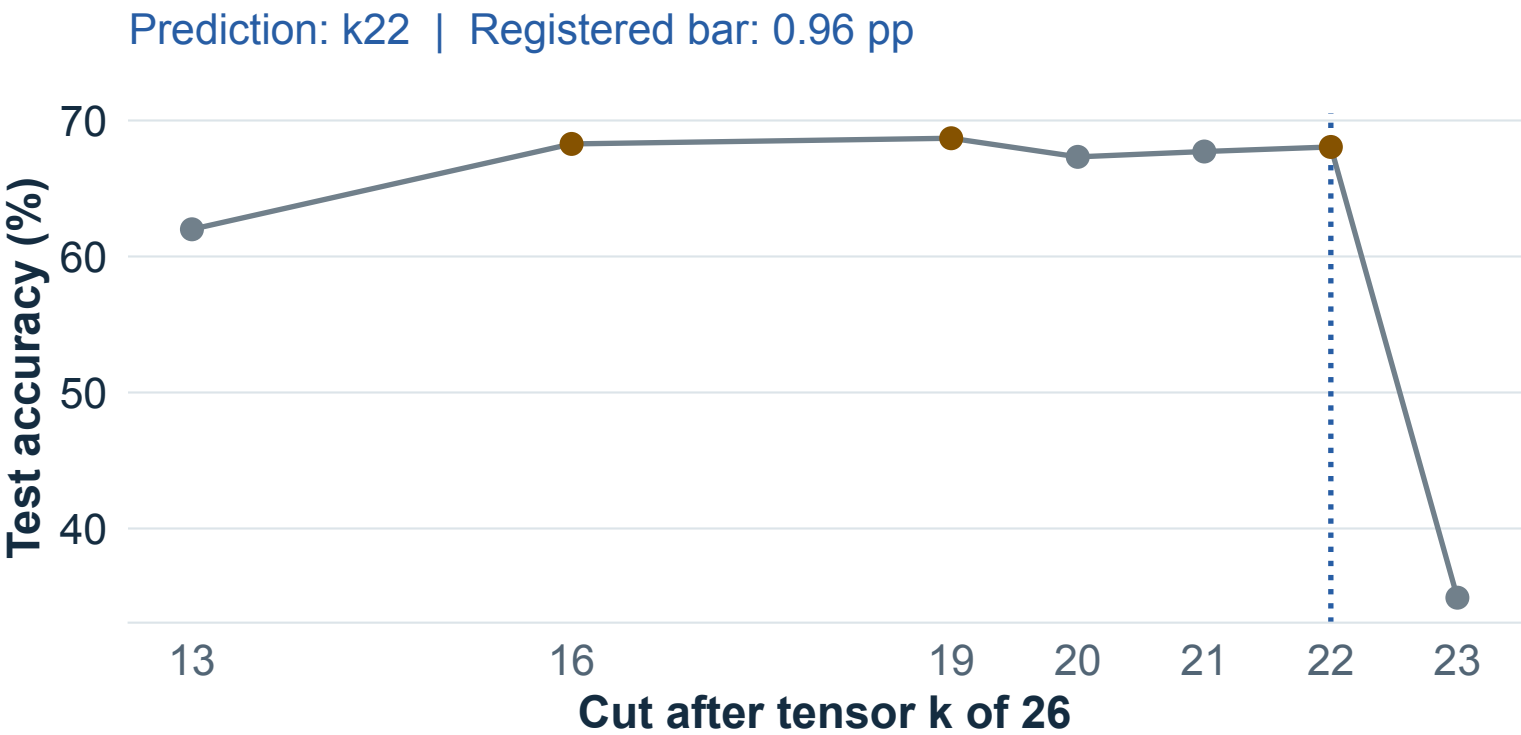
Test whether the best cut sits before bn8.

A vs B

VGG cut ladder: tensor boundary vs balance accounts

WHY

Separate explanations that overlapped on ResNet.



RESULT

TIED

Peak set: 16, 19, 22

Best mean: cut 19

Scalar: 35.45%

Layerwise: 66.28%

WHY THIS RESULT?

The registered separation bar leaves cut 22 tied with rivals.

MetaOptimize loses to scheduled SGD

FAIL

GOAL

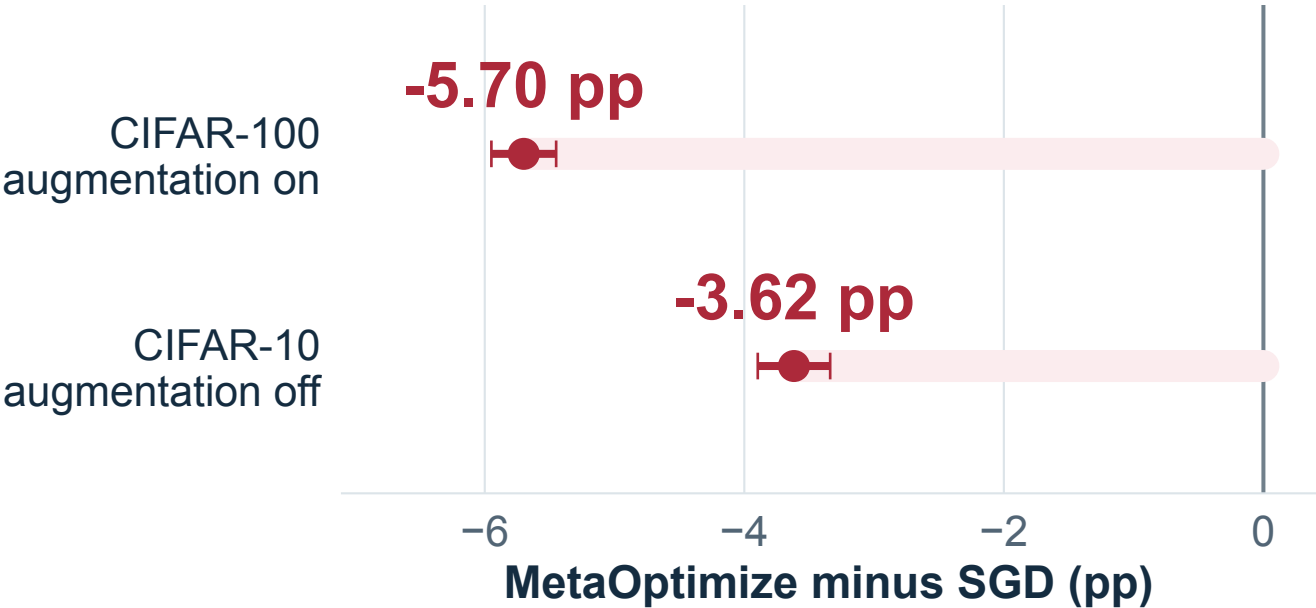
Beat a tuned non-meta optimizer.

A vs B

MetaOptimize vs SGD + momentum + cosine schedule

WHY

Check whether the method offers a practical accuracy win.



RESULT

SGD WINS

77.59 vs 71.89

89.62 vs 86.00

CIFAR-100 / CIFAR-10

Limited method search

WHY FAIL?

Scheduled SGD has higher endpoint accuracy.

Baseline comparisons: full register 1

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT014

CORRECTED

Locate a tuned non-meta baseline

A vs B: AdamW cosine across seven peak learning rates

Cosine peaks near 3e-3; earlier 1e-3 peak was wrong.

The earlier reference understated the scheduled baseline.

MT015

SUCCESS

Test whether scheduling explains the baseline gap

A vs B: Fixed AdamW, MetaOptimize, AdamW cosine, SGDm cosine

Meta beats fixed AdamW by 1.551 pp; scheduled SGD wins by 1.807 pp.

The scheduled denominator changes the method comparison.

MT016

SUCCESS

Validate the experimental harness

A vs B: SGD momentum cosine against a registered accuracy band

SGD reaches the expected band; the best primary cell is 95.124%.

A known reference recipe reproduces expected accuracy.

MT017

FAIL

Close the baseline deficit with longer training

A vs B: Matched recipe families at 100, 300, 600 epochs

A roughly two-point deficit persists at 300 and 600 epochs.

More training did not close this configured baseline deficit.

Baseline comparisons: full register 2

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT018

SUCCESS

Replicate baseline superiority across architecture sizes

A vs B: Cosine and best available meta arms on R34/R50

Cosine remains stronger on R34 and R50; R34 pooling is selection-inflated.

The baseline advantage survives the tested architecture changes.

MT019

OPEN

Audit comparison fairness

A vs B: Budget, tuning allocation, augmentation, and paper configuration

Unaugmented C10 measured; C100 and tuning still limited.

Two method alpha0 cells do not bracket its unaugmented optimum.

MT020

SUCCESS

Test the unaugmented baseline deficit

A vs B: Seven SGD rates versus two MetaOptimize initializations

Unaugmented SGD beats the better registered meta cell by 3.617 pp.

The registered CIFAR-10 deficit survives without augmentation.

Step-size granularity: full register 1

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT026

SUCCESS

Measure a partitioning benefit

A vs B: Scalar and partitioned arms with shared versus tuned steps

Tuning greatly shrinks the scalar-layerwise gap; a small residual remains.

A scoped partition benefit survives tuning controls.

MT027

SUCCESS

Test tolerance to excessive meta steps

A vs B: Each partition’s loss above its selected meta-step peak

Scalar loses more under excessive meta steps; the old nodewise slope is stale.

The ordering survives the registered primary-metric reanalysis.

MT028

MIXED

Resolve accuracy differences among tuned partitions

A vs B: Layerwise, six-block, nodewise and weightwise at selected meta steps

Small coarse differences are batch-sensitive; weightwise trails.

The old 0.254-point ranking lacked a balanced batch comparison.

MT029

FAIL

Recover weightwise accuracy by tuning

A vs B: wm9 weightwise meta-step ladder versus registered 92.2-92.9 band

wm9 peaks at 91.015%; every run misses the registered recovery band.

The measured interior plateau stays below the recovery band.

Colors judge the stated goal. Blue marks a corrected historical claim or measurement.

Step-size granularity: full register 2

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT030

MIXED

Establish a base-optimizer interaction

A vs B: Granularity under SGDm, AdamW, and Lion at two initializations

Optimizer interaction changes with initialization; not universal.

AdamW reverses at proper initialization; Lion also contradicts the mechanism.

MT031

SUCCESS

Explain catastrophic weightwise collapse

A vs B: Lion versus Adam meta updates and beta guard

Guarding removes catastrophic collapse; a sizable weightwise accuracy deficit remains.

The guard fixes collapse but not the measured accuracy loss.

MT033

FAIL

Flatten the fully tuned granularity ladder

A vs B: Best observed meta step for all five partition scales

Completed wm9/tw0 refute a flat ladder; the weightwise rung stays lower.

The finest partition fails to recover through the tested tuning.

MT034

FAIL

Validate shared-step attribution to partition choice

A vs B: Nodewise minus layerwise at common versus individually tuned steps

The contrast changes from +1.138 to -0.360 pp after tuning.

A single shared meta step confounds the attribution.

Training duration: full register 1

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT039

MIXED

Establish an optimal-granularity reversal over time

A vs B: Layerwise versus nodewise through epochs 10-40

The reversal holds at shared $ms=1e-3$; tuning reverses the trajectory.

The observation is real but the optimum interpretation does not survive.

MT040

MIXED

Generalize reversal shape and crossing time

A vs B: Forty-one observed configuration cells across models and datasets

Rising differences occur in 24/29 eligible cells; crossing time does not generalize.

The direction is common within a stratum; timing varies.

MT041

FAIL

Preserve the reversal after meta-step tuning

A vs B: Layerwise at $1e-4$ versus nodewise at $3e-4$

Nodewise starts ahead and declines to -0.360 pp; original reversal fails.

Registered T-B confirms the counterprediction, not the original headline.

Initial-rate robustness: full register 1

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT047

SUCCESS

Measure fixed-step fragility

A vs B: Constant AdamW across seven learning-rate cells

Small fixed rates undertrain; lr=0.1 collapses in all three runs.

Fixed-step performance is strongly sensitive to the chosen rate.

MT048

MIXED

Test robustness across initial learning rates

A vs B: MetaOptimize across the same seven initializations

Low-alpha0 accuracy stays near 91.6%; the highest-rate cell is not robust.

The low-end advantage holds; whole-grid flatness does not.

MT049

SUCCESS

Improve robustness over scheduled training

A vs B: Equal-weight grid means for meta and cosine

Primary grid-mean advantage is 3.159 pp; six-cell sensitivity gives 4.680 pp.

The advantage survives both stated grid reductions.

MT050

SUCCESS

Test robustness at a longer budget

A vs B: 100 versus 300 epochs at four grid extremes

The registered band comparison was stable on the matched extreme cells.

The selected long-budget follow-up did not reverse the registered reading.

Initial-rate robustness: full register 2

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT051

CORRECTED

Audit threshold-based robustness claims

A vs B: Three reductions of the same robustness runs

Three threshold-width verdicts reversed and were withdrawn.

Shared-grid truncation and tiny threshold crossings changed the apparent width.

Model and dataset scope: full register 1

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT057

MIXED

Transfer the granularity gain across model sizes

A vs B: Scalar and layerwise on R10/R18/R34/R50

The gap shrinks with scale and reverses for R50 at small alpha0.

The gain is strongly model-dependent at the shared meta step.

MT058

MIXED

Reproduce the ResNet-50 layerwise deficit

A vs B: R50 scalar versus layerwise at two initializations

R50 layerwise loses 1.356 pp at 1e-6 but gains 1.130 at 1e-3.

Layerwise loses only at the smaller tested initialization.

MT059

SUCCESS

Transfer granularity gains to CIFAR-100

A vs B: Scalar, six-block, layerwise at two initializations

Large gaps repeat; 6-block/layerwise order changes.

The gap transfers while the partition ordering is dataset-dependent.

MT060

CORRECTED

Audit the claimed cost of finer partitions

A vs B: Correct six-block, layerwise, nodewise, weightwise labels

Six-block was mislabeled finer; nodewise beats layerwise and weightwise loses.

The claimed 17.2-point cost compared a coarser partition.

Model and dataset scope: full register 2

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT061

MIXED

Separate startup delay from partition performance

A vs B: Initialization ladders at 20 and 100 epochs

Startup effects are large; 100-epoch differences remain at the tested shared step.

The 20-epoch design also changes pooling and has one seed.

MT062

FAIL

Erase the granularity gap with training time

A vs B: Windows at epochs 100, 205, 300 on identical trajectories

Granularity remains near 3.4 pp; shrink benefits attenuate strongly.

The longer shared-step trajectory does not erase the partition gap.

MT063

CORRECTED

Verify pooling inversion and scalar crossover

A vs B: Reaudit horizon tables using matched primary windows

Corrected cells show attenuation without the claimed inversion.

Best-test selection, mixed initialization, and mislabeled operators created the headline.

Colors judge the stated goal. Blue marks a corrected historical claim or measurement.

Proposed fixes: full register 1

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT069

FAIL

Beat the tuned baseline with a cosine prior

A vs B: Idea-1 meta arms and their available controls

Idea 1 missed its stopping-rule baseline; its exact schedule remains unidentified.

Accuracy did not clear the registered baseline requirement.

MT070

FAIL

Choose partitions using tensor correlation

A vs B: Measured adjacent versus random tensor sign agreement

Powered tensor-correlation contrast: null; direction abandoned.

The signal needed by the proposed partition rule was absent.

MT071

CORRECTED

Find a tunable shrinkage strength

A vs B: Per-step M0 shrink across a lambda ladder

The operator saturates; the alleged tunable pooling interpretation was withdrawn.

Per-step application turns most tested lambdas into near-full shrinkage.

MT072

FAIL

Close the weightwise deficit with pooling

A vs B: Shrink, additive, and zpool weightwise controls

zpool rescues weightwise accuracy but still misses the stronger coarse partitions.

A rescue exists, but it does not close the competitive deficit.

Proposed fixes: full register 2

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT073

SUCCESS

Find an interior additive optimum

A vs B: Layerwise additive r curve against both endpoints

Interior benefit survives meta-step tuning: +0.821 pp.

The later joint sweep defeats the pure meta-step explanation.

MT074

CORRECTED

Validate the additive pooling interpretation

A vs B: Additive and true-pooling endpoint controls across granularities

Pooling claim withdrawn; pure-rescaling account refuted.

g4m overturns one explanation without validating the original operator interpretation.

MT075

FAIL

Improve accuracy through true full pooling

A vs B: zpool r=0 versus plain layerwise

True full pooling reduces accuracy by about three points.

The earlier positive endpoint was an additive arm, not true pooling.

MT076

MIXED

Establish a competitive tuned pooling method

A vs B: Additive and plain controls with meta-step tuning

Older partial tuning favored plain; g4m later preserves a local additive advantage.

The original broad domination claim did not test the joint sweep.

Proposed fixes: full register 3

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT077

FAIL

Preserve the additive optimum at proper initialization

A vs B: Additive r curves at $\alpha_0=1e-6$ and $1e-3$

Most interior gain disappears; the whole curve does not become flat.

The remaining local improvement misses the stated materiality threshold.

MT078

MIXED

Transfer the additive optimum

A vs B: r curves across architecture, dataset, and initialization

R10/R34 small- α_0 optima exist; C100 fails; R50 follow-up remains unresolved.

Transfer is conditional; the old “nowhere” claim is false.

MT079

MIXED

Improve good partitions with meta-gradient pooling

A vs B: zpool r sweeps for layerwise and weightwise

Weightwise is rescued; layerwise shows no material interior improvement.

The benefit is a fine-partition rescue, with interior step-size confounding.

MT080

OPEN

Identify a scaling law for optimal r

A vs B: R10/R18/R34 argmaxes versus parameter count

Batch noise and coarse grid prevent a scaling law.

Cross-family peak margins are below the measured batch floor.

Meta-gradient measurements: full register 1

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT086

CORRECTED

Check the claimed excess sign agreement

A vs B: Layerwise and weightwise agreement vs their own independence nulls

53.1% belongs to 62 groups; its independence expectation is 55.07%.

The published sentence combined the wrong arm and wrong null.

MT087

FAIL

Test inverse-square-root noise averaging

A vs B: Measured drift slopes vs the square-root averaging prediction

Direction rejects square-root scaling; magnitudes are censored.

The measured scaling does not match the registered expectation.

MT088

SUCCESS

Detect correlation after step sizes adapt

A vs B: Frozen vs free beta at the same R18 configuration

Adapted correlation: 8.458e-8; replication: 8.888e-8.

The adapted estimate exceeds its own resolution floor.

MT089

SUCCESS

Measure effective independence across model and dataset families

A vs B: Free vs frozen beta across four architecture/dataset combinations

Effective independence: 16-55%; SGDm, epochs 10-20.

The fraction remains below independence across all four tested families.

Colors judge the stated goal. Blue marks a corrected historical claim or measurement.

Meta-gradient measurements: full register 2

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT090

FAIL

Preserve effective independence in later training windows

A vs B: Epochs 10-20 vs 20-40 at each granularity

Nodewise and weightwise decline; layerwise change remains unresolved.

The effective-independence fraction changes substantially in the finer partitions.

MT091

SUCCESS

Keep effective-independence measurements robust to guard changes

A vs B: Published guard boxes vs measured box-free configurations

Box removal changes N_eff/m only -0.006 to +0.024 across four families.

All shifts stay within the registered measurement tolerance.

MT092

SUCCESS

Test gradual adaptation of the correlation profile

A vs B: Frozen beta vs two valid meta-step settings

N_eff/m rises 0.042 → 0.129 → 0.481 as meta-step increases.

The intermediate setting lands between the two endpoint profiles.

MT093

SUCCESS

Verify short-range correlation beyond estimator artifacts

A vs B: Corrected scale profiles vs exchangeable and independence models

Frozen-beta profiles fall 16.6-264.6× across the four tested families.

The falling profiles survive the corrected independence floor.

Meta-gradient measurements: full register 3

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT094

FAIL

Test whether row means represent within-row structure

A vs B: Observed row structure vs size-preserving within-tensor regrouping nulls

Most marginal-sign variation remains within rows; clean row means explain very little.

The measured structure is poorly represented by the row means.

MT095

MIXED

Find reproducible architecture-linked row structure

A vs B: Anomalous vs clean runs, then row identity across seeds

Anomalous runs have real structure, but the responsible row identities do not reproduce.

Within-run structure is strong; cross-seed identity stays near its null.

MT096

FAIL

Detect a coherent convolution-filter interaction

A vs B: Output-input channel interaction vs a within-tensor permutation null

All 50 clean arms miss the 1.10 interaction threshold.

Measured interaction ratios span only 0.811-1.041.

MT097

MIXED

Locate a stable scale of marginal sign structure

A vs B: Field block-size curves across trajectory and meta-step settings

Peaks cluster within one rung; four clean arms remain bounded by the ladder edge.

Location is stable at tested resolution, but not fully bracketed.

Colors judge the stated goal. Blue marks a corrected historical claim or measurement.

Meta-gradient measurements: full register 4

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT098

FAIL

Predict useful partitions from field measurements

A vs B: Field-preferred scales and effective independence vs measured accuracy

The scale bridge fails; later cc1 shows anti-concordance and dissociation.

Better field readings do not consistently produce better accuracy.

MT099

OPEN

Explain the late-training correlation rebound

A vs B: Original clipping vs box-free repeats and beta-velocity evidence

Rebound persists after unclipping; its cause remains unexplained.

Both tested mechanisms fail; no third mechanism was registered.

MT100

SUCCESS

Verify the exponent contrast with independent estimators

A vs B: Raw agreement, debiased agreement, and variance on matched runs

Corrected instruments agree near 0.18; the raw 0.343 gap was overstated.

The direction survives two instruments after removing their bias mismatch.

MT102

SUCCESS

Test span against time and meta-step explanations

A vs B: Matched time or meta-step with different adaptation spans

Span separates the resolved orderings; two of eight cells remain undecided.

sp8 breaks the earlier time-versus-span confound.

Colors judge the stated goal. Blue marks a corrected historical claim or measurement.

Meta-gradient measurements: full register 5

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT103

FAIL

Preserve partition ordering across base optimizers

A vs B: SGDm vs AdamW with the same three granularities

At this untuned 20-epoch setting, AdamW changes the ordering.

The smallest fraction moves from nodewise to weightwise.

Measurement checks: full register 1

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT108

CORRECTED

Compare equal metric windows across training budgets

A vs B: Legacy last-20 averages vs explicit last-five accuracy

The 20-epoch gap shrinks from 1.67 to 0.07 pp under matched windows.

Legacy plateau averages the entire 20-epoch learning curve.

MT109

CORRECTED

Build response surfaces from completed runs

A vs B: Requested vs completed epochs in each response-surface arm

Twelve truncated blk6/nodewise runs distorted the response surface.

Short runs depressed selected arms and could reverse curve shape.

MT110

CORRECTED

Evaluate weightwise using the correct training budget

A vs B: 20-epoch weightwise records vs the matched 100-epoch anchor

Correct plateau5 is 82.086% at 20 epochs and 91.308% at 100 epochs.

The discarded-arm justification mixed both budget and metric.

MT111

CORRECTED

Verify derived probe fields against raw quantities

A vs B: Stored dimensions, capped counts, spread fields, and collapse flags

Count, spread, and collapse definitions introduced four distinct analysis failures.

Derived columns silently encoded assumptions that raw measurements contradicted.

Colors judge the stated goal. Blue marks a corrected historical claim or measurement.

Measurement checks: full register 2

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT112

CORRECTED

Report statistics only after runs finish

A vs B: The same probe at 89 vs 100 records

Reading early changed the reported steady-window excess by 49%.

The window was defined relative to the records currently present.

MT113

CORRECTED

Measure guard occupancy at coordinate resolution

A vs B: Per-tensor mean betas vs actual coordinate clip counts

A reported zero ceiling occupancy hid binding on 95.4% of records.

Tensor averages concealed individual coordinates touching the guard.

MT114

FAIL

Predict future guard contact from current velocity

A vs B: Projected guard contact vs the observed continuation

One seed reaccelerates; contact occurs 12.4 epochs earlier than predicted.

A declining observed velocity did not remain declining.

MT115

CORRECTED

Detect guard contact in every seed

A vs B: Pooled occupancy vs each run's occupancy

A pooled 8.03% concealed one seed at 24.07% occupancy.

Pooling hid a seed pinned through 96.3% of its last quarter.

Measurement checks: full register 3

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT116

CORRECTED

Require every ranked arm to pass validity gates

A vs B: Argmin claims with all-rung interpretability checks
12/25 old cells uninterpretable; scoped bo7 result later resolved.

The selected minimum inherits every ranked rung's validity requirements.

MT117

CORRECTED

Verify claimed configuration matches before submission

A vs B: Declared byte matches vs actual arguments and queue selections

Wrong head: 6 jobs. Separate mistaken cancellation: 50 jobs.

Headers and submission recency were unreliable proxies for intended configuration.

MT118

CORRECTED

Make submission throttling respond to usable capacity

A vs B: FairShare-based blocking vs measured decay and actual queue reasons

FairShare barely moved; 93 of 94 pending jobs were quota-bound.

The old floor blocked submissions without addressing the active limit.

MT119

CORRECTED

Identify which registered tests remain unfinished

A vs B: Historical outage list vs landed runs and later decisions

Most batches landed; cp9 and free-coarse field remain unverified.

The outage snapshot is superseded by subsequent runs and explicit closures.

Colors judge the stated goal. Blue marks a corrected historical claim or measurement.

Measurement checks: full register 4

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT121

FAIL

Ensure the requested measurement instrument actually runs

A vs B: Installed probe patch vs enabled runtime instrument

bf8 completed 12 jobs but recorded zero effective-independence data.

The launcher set probe stride but omitted the instrument switch.

MT122

CORRECTED

Identify which guard caused the failed validity gate

A vs B: Floor and ceiling occupancy with both denominators

bf8 freed the floor; the remaining bind was a sparse ceiling event.

The gate caught binding, but the prose named the wrong wall.

MT123

CORRECTED

Measure the true uncited-run backlog

A vs B: Full-document census vs an artificially restricted citation search

~290-run backlog withdrawn; uncited smoke tests retired.

Omitting the main evidence documents manufactured most apparent orphans.

MT124

CORRECTED

Check headline contrasts for matched experimental conditions

A vs B: Published contrasts vs full-axis reconstruction

13/56 audited pairs contaminated; 8 verdicts flipped.

Incomplete matching changed both estimates and their interpretations.

Mechanism and isolation: full register 1

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT132

FAIL

Explain r benefit through meta stepsize rescaling

A vs B: Interior r versus endpoints after tuning each meta stepsize

Tuned interior gain +0.821 pp refutes simple rescaling.

Gain exceeds the registered 0.50 pp bar.

MT133

FAIL

Improve layerwise learning using two-level shrinkage

A vs B: Partial shrinkage versus plain layerwise at two initial stepsizes

Best partial shrinkage trails layerwise by 0.253 and 0.819 pp.

Every partial-shrinkage setting loses at both initial stepsizes.

MT134

FAIL

Recover pooling gains without tuning a shrinkage dial

A vs B: Empirical Bayes shrinkage versus identity and tuned additive controls

Primary lift +0.0413 pp misses the 1.16 pp success bar.

Lift falls within the registered failure region.

MT135

OPEN

Replicate an interior r optimum on ResNet-50

A vs B: Six additive r values at two initial stepsizes

Both strata unresolved; SPLIT does not establish transfer.

Neither registered peak confirmation nor flat-curve refutation fires.

Mechanism and isolation: full register 2

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT136

SUCCESS

Replicate the granularity gap on Tiny-ImageNet

A vs B: Scalar, layerwise, nodewise, plus CIFAR-10 harness controls

Layerwise exceeds scalar by 40.9753 pp on Tiny-ImageNet.

The registered dataset replication passes.

MT137

FAIL

Explain the granularity gap through the classifier head

A vs B: Head split versus stem, midpoint, blocks, and endpoint controls

Head captures 4.12%; midpoint captures 21.65%; head specificity is negative.

An arbitrary midpoint control outperforms the head split.

MT138

OPEN

Determine whether the additive peak is Lion-specific

A vs B: Adam and Lion additive r ladders with internal endpoints

All 24 runs completed; Adam grid remains truncated near its best interior rung.

Unsampled r below 0.03 prevents the registered conclusion.

MT139

MIXED

Explain accuracy through group count or size balance

A vs B: Balanced count ladder and fixed-count mirrored two-group partitions

Mirrored cuts have identical tensor-count entropy but differ by 16.5393 pp.

Tensor-count entropy fails; count alone cannot explain fixed-count differences.

Mechanism and isolation: full register 3

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT140

OPEN

Test granularity effects on ImageNet-489

A vs B: Scalar, six blocks, layerwise, and nodewise at common epoch 80

Scalar and blocks fail the learned-floor gate; nodewise improves over layerwise.

Registered BLOCK and GRAN gates refuse near-floor scalar and block arms.

MT141

SUCCESS

Locate the best two-group cut

A vs B: Thirteen cut and anchor arms at 100 epochs

Sampled peak: k49 at 55.4473%; balance account survives.

Cut localization succeeds, but carrier and parameter-balance accounts coincide.

MT142

SUCCESS

Identify the tensor carrying the first cliff

A vs B: Adjacent cuts k49, k50, k51, k52 with scalar/layerwise anchors

Moving BN scale 50 into the coarse group lowers accuracy by 24.8340 pp.

A 512-parameter scale dominates the local 100-epoch cliff.

MT143

FAIL

Explain the cliff through dependence on beta clamp

A vs B: Clamped and released boxes crossed with cuts k49 and k50

Released cliff 24.7673 pp; clamped cliff 25.1627 pp; interaction unresolved.

The cliff survives release; clamp dependence is refuted.

Mechanism and isolation: full register 4

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT144

OPEN

Resolve scalar clamp dependence using existing runs

A vs B: Released scalar contrasts versus canonical contrasts across archived batches

Old audits open; later controls resolve headline endpoint claims.

Middle rung remains constrained; blockwise remains unaudited.

MT145

SUCCESS

Identify the tensor carrying the second cliff

A vs B: Adjacent cuts k52 through k55

Moving shortcut BN scale 53 lowers accuracy by 15.8080 pp at 100 epochs.

One 512-parameter scale carries the second measured cliff.

MT146

SUCCESS

Test cliff persistence beyond 100 epochs

A vs B: Cuts k49 and k50 within the same runs at 100/772 epochs

Cliff persists at 19.3427 pp and shrinks by 5.6747 pp, as predicted.

Persistence and predicted shrinkage both pass their registered bars.

MT147

MIXED

Preserve peak and predict longer-horizon ranks

A vs B: Five sampled cuts at 100/772 epochs on fresh seeds

Peak remains k49; none of three forecast rank swaps occurs.

Location prediction succeeds; its curve model fails.

Colors judge the stated goal. Blue marks a corrected historical claim or measurement.

Mechanism and isolation: full register 5

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT148

FAIL

Explain long-horizon curves by changing only rate

A vs B: Windowed k50/k52 approaches against an exponential shape null

k52 differs in shape; the decisive rho(0.90) separation is 1.79 registered bars.

The exponential shape account fails for k52, not k50.

MT149

FAIL

Explain headline gap through an alpha0 confound

A vs B: Matched scalar/block/layerwise contrasts at two initial stepsizes within batches

Ordering persists; the alleged 24.6 pp swing does not reproduce in matched batches.

A smaller blockwise interaction exists; the alleged confound magnitude fails.

MT150

MIXED

Localize peak and resolve tensor-class effects

A vs B: Cuts k45 through k50 at 772 epochs on fresh seeds

Peak remains k49; observed class signs are perfectly aliased with index modulo three.

Localization succeeds; tensor-class mechanism remains unidentifiable.

MT151

FAIL

Locate the ImageNet transition using group count

A vs B: Six balanced counts plus a second eight-group partition

Transition is distributed; at eight groups, capture changes from 0.0080 to 0.1497.

Identical group counts yield different captures; no registered cliff fires.

Colors judge the stated goal. Blue marks a corrected historical claim or measurement.

Mechanism and isolation: full register 6

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT152

OPEN

Separate tensor identity from ordinal position

A vs B: Named-set permutations and tensor swaps versus equivalent prefix controls

Single-tensor additivity fails; identity vs order stays open.

A defective registered branch cannot support the intended mechanism.

MT153

MIXED

Separate contiguity effects from coarse parameter mass

A vs B: Co-grouping, holes, and an exact 3,956,288-parameter mass match

Coarse-mass account fails; contiguity remains confounded.

Mass is refuted; contiguity and a membership rival remain inseparable.

MT154

CORRECTED

Check the claimed privilege of conv2 tensors

A vs B: Re-examine the existing tensor-membership rival

conv2 privilege fails; the substantive rival remains unresolved.

A naming error does not eliminate the competing account.

MT155

FAIL

Beat a tuned non-meta CIFAR-100 baseline

A vs B: MetaOptimize versus six SGD learning rates and a 200-epoch control

Best SGD exceeds MetaOptimize by 5.6993 pp; 200-epoch SGD reaches 78.8280%.

Every tested SGD rung beats the tested MetaOptimize arm.

Colors judge the stated goal. Blue marks a corrected historical claim or measurement.

Mechanism and isolation: full register 7

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT156

MIXED

Explain cliff amplification through reduction composition

A vs B: Raw tensor sums versus per-tensor numel normalization at three cuts

One clean pair changes accuracy; other effects include failure.

Only k49 avoids the failed-arm floor in both reductions.

MT157

MIXED

Retain granularity benefits after independent stepsize tuning

A vs B: Scalar/layerwise meta-stepsize ladders at two initial stepsizes

At alpha0=1e-3, gap remains 35.2907 pp; alpha0=1e-6 optimum lies at ladder edge.

One stratum survives tuning; the other is unresolved.

MT158

CORRECTED

Verify coverage of the low-meta-stepsize stratum

A vs B: Corpus census against the prior prose inventory

The stratum contained 35 rows, two alpha0 levels, and scalar/layerwise controls.

Earlier counts and the claimed alpha0 alias were stale.

MT159

FAIL

Explain capture peak through between-group sign disagreement

A vs B: Reconstructed window disagreement versus the registered cut-position curve

Disagreement peaks elsewhere; correlation with capture is -0.0455.

The disagreement maximum does not align with the accuracy peak.

Colors judge the stated goal. Blue marks a corrected historical claim or measurement.

Mechanism and isolation: full register 8

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT160

OPEN

Test element-count majority as the scalar sign driver

A vs B: Proposed element fraction versus the actual scalar probe schema

No archived element-count statistic; scalar aggregate only.

The proposed measure is tautological and cannot test element majority.

MT161

FAIL

Find convolution or head terms driving scalar pinning

A vs B: Per-tensor Lion terms along scalar and layerwise trajectories

Three BN scales set the sign on every pinned record.

The named hypothesis is refuted; a different carrier set is identified.

MT162

MIXED

Rescue learning and free the shared complement

A vs B: Three-carrier isolation, depth-confounded control, one-carrier isolation, and anchors

ISO rescue +46.93 pp; registered freedom gate remains unmet.

Accuracy rescues, while the full registered dynamics conjunction fails.

MT163

SUCCESS

Separate carrier isolation from matched depth controls

A vs B: Carrier triple versus matched non-carriers, scale-only pair, singleton, and anchors

Carrier isolation beats the matched triple by 46.5233 pp; magnitude remains confounded.

Matched non-carriers fail to rescue under the registered identity test.

Colors judge the stated goal. Blue marks a corrected historical claim or measurement.

Mechanism and isolation: full register 9

Each card: GOAL / A vs B / RESULT / WHY THIS OUTCOME

MT164

SUCCESS

Replicate the granularity gap outside ResNet

A vs B: Scalar versus layerwise on VGG11 with BatchNorm

VGG layerwise exceeds scalar by 31.2687 pp; later cvi1 tests isolation.

The gap replicates on a second architecture family.

MT165

OPEN

Retain rescue after the complement meets pinning gate

A vs B: Same-run isolation and anchor plateaus at 100/250 epochs

Gap retention 99.77%; registered exact-pinning gate unmet.

Accuracy persists, but the scorer stops before its primary retention branch.

MT166

SUCCESS

Replicate carrier-specific isolation on VGG

A vs B: bn8 isolation versus its matched bn7 control and anchors

ISO-control +31.30 pp; magnitude and horizon limits remain.

The nominated carrier rescues; its matched control stays at the floor.

CVK2

OPEN

Test the predicted VGG cut location

A vs B: Seven cuts plus scalar and layerwise controls

Cuts 16, 19, 22 share the peak set; prediction unresolved.

The registered separation bar leaves cut 22 tied with rivals.

Data, scope and reproducibility

Historical coverage	110 question and audit entries across 9 areas
New follow-up	CVK2: 27 jobs, 9 configurations, 3 seeds
Accuracy metric	Mean accuracy over the last 5 completed epochs
Uncertainty	Seed SE where available; no invented error bars
Source hierarchy	Raw logs, registered scorers, corrected research records
Open results	Unresolved, declined and withdrawn tests remain labelled
Supporting files	Complete register, figure index and numeric tables in the ZIP
Inventory limit	177 other historical GPU jobs lie outside the run ledger

This work was performed using the compute resources from the Academic Leiden Interdisciplinary Cluster Environment (ALICE) provided by Leiden University.